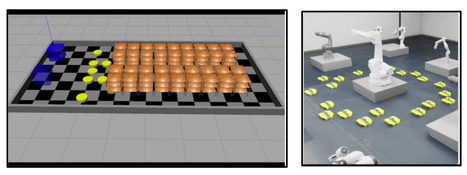
Model-Based Diffusion Optimal Control for Multi-Robot Motion Planning

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https://github.com/hhhhzl/mdoc

1 Motivations



Multi-Robot Motion Planning (MRMP) requires:

- Dynamic feasibility
- Collision-free trajectories
- Scalability

Why the problem is hard?

- High dimensional space (joint trajectory space)
  - Classical sampling-based methods are **slow and sample-inefficient**.
  - Optimization-based methods require solving **large non-convex constrained programs.**
- Coupled safety constraints
  - Dynamics, obstacles, and robot-robot collisions **must be satisfied** simultaneously.

2 Related Work & Their issues

$$\tau^\star \in \arg\min_{\tau^{1:T}} \quad \mathcal{J}(\tau^{1:T})$$
$$\text{s.t.} \quad s^{t+1} = f(s^t, u^t), \quad g^t(s^t, u^t) \le 0,$$

Multi-Robot Motion Planning with Diffusion Models (MMD) [Shaoul et al. 2025]

- Learning-based
- Strength in generating multi modality trajectory solutions
- Issues
  - **Need pre-generated or pre-collected data (massive!)**
  - **Pretrain a network (model-free)**
  - **Dynamics?? Constraints??**

We Need:

(1) Model-Based Approach!!

(2) Constraints-aware Trajectory!!

(3) Multi-robot motion planning !!

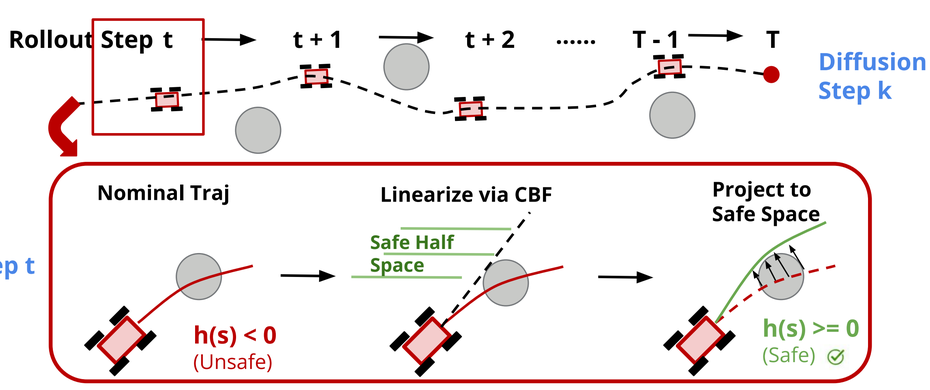
Diffusion Steps in Loop

Sample From Gaussian Kernel	$\tilde{\tau} \sim \Omega_k := \mathcal{N}\left(\frac{\tau_k}{\sqrt{\bar{\alpha}_k}}, \left(\frac{1}{\bar{\alpha}_k} - 1\right)\mathbf{I}\right),$	Sampling on control input
Rollout Dynamics	$s^{t+1} = f(s^t, u^t),$	Getting Batch Nominal Trajectories
Monte Carlo Reweighted	$\bar{\tau}_k = \frac{\sum_{m=1}^{M} p_1(\tilde{\tau}_{k,m})\,\tilde{\tau}_{k,m}}{\sum_{m=1}^{M} p_1(\tilde{\tau}_{k,m})},$	Evaluated based on cost or reward
Diffusion Score Compute	$\hat{\mathbf{S}}_k \approx -\frac{\tau_k}{1-\bar{\alpha}_k} + \frac{\sqrt{\bar{\alpha}_k}}{1-\bar{\alpha}_k}\,\bar{\tau}_k.$	Score is analytically calculated !!! Score = Gradient of Log Density (learned in model-free approaches)
Monte Carlo Score Ascent	${}^{k-1}\tau^{1:H} = \frac{1}{\sqrt{\alpha_k}}\left({}^{k}\tau^{1:H} + (1-\bar{\alpha}_k)\,\hat{\mathbf{S}}_k\right),$	Trajectory update

3 MDOC: Single robot planner that handles constraints

Diffusion Steps in Loop

Sample From Gaussian Kernel → Rollout Dynamics with Per-Step CBF ✓ → Monte Carlo Reweighted → Diffusion Score Compute → Monte Carlo Score Ascent

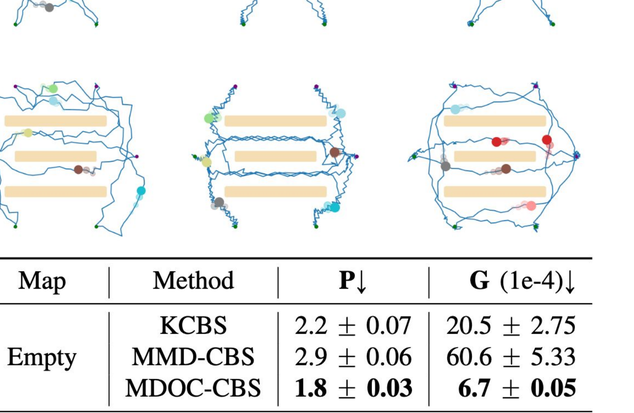


Safety is implemented during the sampling, before the score is computed.
Therefore, the score is constraints-aware.

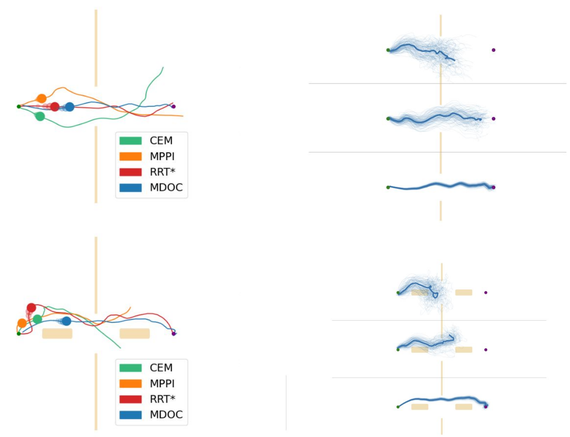
4 MDOC-CBS: Scalable Multi-robot Coordination with MDOC as the low-level planner

High-Level Idea:

- Detect the inter-robot conflicts via Conflict-Based Search (CBS) [Sharon et al. 2015]
- Expand to a Constraint Tree (CT) and add a time-indexed sphere constraint to one conflicted robot
- Replan such robot for both inter-robot conflicts and obstacle constraints with MDOC
- Repeat until **conflict-free**



5 MDOC: Results

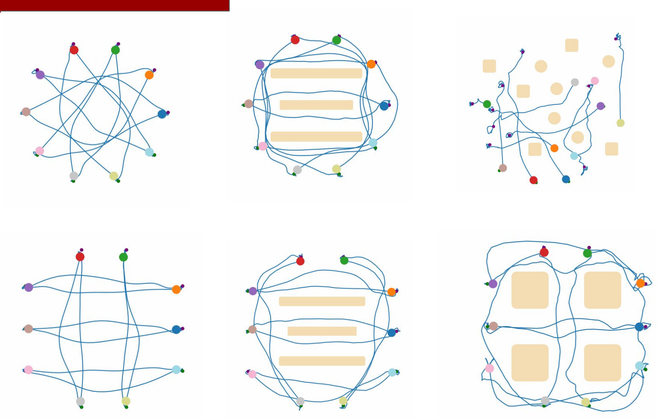


	CEM	MPPI	RRT*	MDOC
Top	2.1 ± 0.2	3.2 ± 0.3	66.1 ± 6.0	**100.0**
Bottom	0.2 ± 0.1	0.4 ± 0.1	41.7 ± 3.4	**100.0**

100% Sample Trajectories Success Rate

Efficient in finding feasible path while remaining collision-free in constrained environment

6 MDOC-CBS: Results



Key Findings
- Higher Success Rate
- **5x – 10x** lower planning time than MMD-CBS
- Shorter and **smoother** Trajectories
- **Scale to 20 robots and 40 robots in larger maps**

Map	Method	P↓	G (1e-4)↓
Empty	KCBS	2.2 ± 0.07	20.5 ± 2.75
	MMD-CBS	2.9 ± 0.06	60.6 ± 5.33
	MDOC-CBS	**1.8 ± 0.03**	**6.7 ± 0.05**
Conveyor	KCBS	**2.7 ± 0.10**	20.7 ± 2.12
	MMD-CBS	3.9 ± 0.11	85.0 ± 5.45
	MDOC-CBS	2.8 ± 0.04	**12.9 ± 1.40**

TABLE II: Comparison of methods in terms of average path length **P** and geometric (Laplacian) smoothness **G** over success trails of all agent numbers, evaluated on the Empty map and Conveyor map in the Circle setup.

Highlights: (1) Training-Free (2) Two Levels of Planning
- (i) Dynamics && Constraint-aware diffusion score
- (ii) CBF per-step fast rollout
- (iii) CBS for large scale MRMP